The mass spectrum of the IIB flux landscape carries no geometric fingerprint

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Abstract

A central hope of the "structure mining" program is that some low-energy observable of a flux vacuum carries a *fingerprint*—information about the flux that generated it, beyond its overall scale—so that the compactification could be pinned down from observables. We test the last candidate: the mass spectrum. On 133 supersymmetric IIB flux vacua (JAXVacua, $h^{1,2} = 2$, natural ISD measure) we compute the degree-0 spectrum—the six eigenvalues of the complex-structure moduli and the axio-dilaton—and measure how predictable each eigenvalue is from the flux. The raw eigenvalues are predictable (ridge-on-quadratic CV $r = 0.50$–0.62, heaviest slot significant at $z = 5.5$), but the *scale-invariant* ratios $m_i^2/m_{\max}^2$ are not ($r < 0.25$). The conclusion: the mass spectrum's *shape* is statistically uniform in flux—the fingerprint is not there—while its *overall scale* is flux-determined through a quadratic form distinct from the D3 tadpole. This closes, in the flux coordinates, the "masses carry the fingerprint" hypothesis, and leaves the constraint (the tadpole) as the only learnable structure found to date.

1 Introduction

The string landscape is $\sim 10^{500}$ vacua, and no experiment distinguishes most of them. The problem of *verification* (does some vacuum match our universe?) is the subject of a companion paper [1]; here we test one specific hypothesis that has guided the search for structure in the landscape: that *the mass spectrum is the fingerprint*—the low-energy observable that carries information about the flux, so that measuring the masses would pin the compactification.

This hope is natural. The superpotential W_0, the string coupling g_s, and the gauge couplings have all been measured to be statistically uniform in the flux coordinates (cross-validated $r < 0.25$) [1]: they are random draws that carry no information about the flux quanta that produced them. The mass spectrum was the one observable not yet measured, and it was the natural candidate for the fingerprint, because it depends on the full geometry through the stabilized moduli rather than on a single closed form.

We compute it, and we find the hope is not borne out.

2 Method

The sample. We use JAXVacua [2] on a bundled $h^{1,2} = 2$ flux EFT ("KS" model). At maximum_degree = 0 (no instanton corrections) the Kähler moduli are flat directions, so the mass matrix has 6 physical eigenvalues: 4 from the two complex-structure moduli and 2 from the axio-dilaton. Critical points are sampled with the natural ISD measure (`isd_mode = ISD−`, Gaussian flux prior); supersymmetric vacua are selected by $DW = 0$ and $\mathrm{Im}\,\tau > 0$. We obtain 133 SUSY vacua from 964 critical points.

The test. For each eigenvalue m_i^2 we measure the cross-validated ridge regression of m_i^2 on the degree-2 monomial basis of the 12 flux coordinates (91 features), reporting the Pearson r between held-out predictions and truth. This is the same joint learnability test used throughout [1]: r near 1 means deterministic structure (the tadpole control), r near 0 means statistically uniform.

Scale vs. shape. A nonzero r can arise in two ways: the *scale* of the spectrum (the overall curvature, set by the potential) or the *shape* (the ratios m_i^2/m_j^2). To separate them we re-run the test on the scale-invariant ratios $m_i^2/m_{\max}^2$. If the shape carries the fingerprint, the ratios are learnable; if only the scale does, the raw values are learnable but the ratios are not.

3 Results

	eigenvalue slot (sorted ascending)					
	0	1	2	3	4	5
raw m_i^2	0.03	0.37	0.50	0.53	0.58	**0.62**
ratio m_i^2/m_5^2	0.09	0.16	0.21	0.15	0.08	—

The raw eigenvalues are learnable: the three heaviest slots cross the $r = 0.5$ threshold, and the heaviest (m_5^2) is significant at $z = 5.5$ against a permutation null (shuffled flux labels; null mean ≈ 0.001, std ≈ 0.11). The scale-invariant ratios are *not* learnable: all $r < 0.25$, inside the null. The heaviest eigenvalue correlates with the D3 tadpole at only $r = 0.12$, so the learned scale is a quadratic form of the flux *distinct* from the tadpole, not the tadpole itself.

4 Conclusion

Remark 1. ***The mass spectrum's shape is statistically uniform in flux; its scale is flux-determined.*** *In the IIB flux corner, on* 133 *SUSY vacua, the degree-*0 *mass eigenvalues carry information about the flux only through their overall scale. The scale-invariant ratios—the part that would constitute a geometric fingerprint—are random ($r < 0.25$). The fingerprint is not in the mass spectrum.*

This closes the "masses carry the fingerprint" hypothesis in the flux coordinates, and completes the pattern: W_0, g_s, the gauge couplings, and now the mass-spectrum shape are all statistically uniform in flux. The only learnable structure found to date is the *constraint*—the D3 tadpole, which is the exact symplectic pairing. The constraint-first program of [1] therefore survives its first decisive falsification attempt: the landscape's navigable structure is what it *forbids*, not what it *selects*.

Scope, named. (i) This is the degree-0 spectrum; the instanton-corrected Kähler-moduli masses remain computationally infeasible and could in principle differ. (ii) "Random" means *in the flux coordinates*; the masses could still carry structure in the moduli coordinates, which we have not tested. (iii) It is a measurement on one model ($h^{1,2} = 2$) at 133 vacua, not a theorem.

References

[1] R. Mitchell, "Probably Approximately Correct String Theory: A Verification Discipline for the Landscape" (companion paper).

[2] A. Dubey, S. Krippendorf, A. Schachner, "JAXVacua" (arXiv:2306.06160).

[3] F. Denef and M. R. Douglas, "Distributions of flux vacua," JHEP **0405** (2004) 072.